Vibration Modelling of Waveguide Structures using the Wave and Finite Element Method

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ABSTRACT: The vibration modelling of waveguide structures is considered. These structures comprise waveguides connected via joints. Traditionally, the wave behaviour of such structures can be analysed if they are simple (beams, rods, etc.). However, if the waveguides are of complicated constructions (truss-like, layered media, etc.), obtaining the wave characteristics might be a formidable task. In this paper, such structures are modelled using a hybrid wave and finite element/finite element (WFE/FE) approach. The waveguides are modelled using the WFE method and thus their wave characteristics are obtained regardless of the complexity of their cross-section. The joints are modelled using standard FE, and the WFE and FE models are coupled to yield the scattering properties of the joints. These are used to describe the behaviour of the structure using relatively small models while providing important information for other applications such as structure-borne sound, statistical energy analysis, etc. Numerical examples are presented to illustrate the approach.

KEY WORDS: wave and finite element, train cross-section, reflection and transmission coefficients, structural networks.

1 INTRODUCTION

Many structures comprise waveguides which are connected via joints into some configuration. The vibrational behaviour of such a structural network can be described through the waves that travel through the waveguides, and the scattering properties of each joint. For simple structures (where waveguides are connected at point joints for example), the wave characteristics of the waveguides can be obtained analytically along with the scattering properties of the joints [1-3]. For complicated structures, analytical solutions might be very difficult to obtain. In this case, the finite element (FE) method is often used. However, FE models become impractical at higher frequencies leading to a multitude of problems (computation speed, accuracy, etc.).

The analysis of waveguide structures has attracted much research. The approaches developed can be categorised into matrix- and wave-based methods. The former uses a matrix formulation to couple the waveguides that constitute the structure [4]. Examples include the dynamic stiffness method [5-7], mobility power-flow analysis [8], the wave scattering approach [9], and the spectral element method [10]. Other researchers presented a matrix formulation with wave scattering [1, 11, 12] to analyse waveguide structures.

Wave propagation methods rely on obtaining the reflection and transmission coefficients of the joints. For thin beams, Cremer et al. [13] calculated the scattering coefficients of joints due to an incident propagating wave. Mace [14] considered the scattering due to an incident evanescent wave, and later studied the properties of the scattering coefficients [15]. Researchers also used wave-based methods to treat three-dimensional frames [9], truss-like structures [16], and structural networks [2, 3]. Wave approaches were also used to calculate the reflection and transmission coefficients of plate/beam junctions [17], bolted joints [18], curved beams [19] and to analyse the vibrational energy in built-up structures [20-22]. Heron [23] used a line impedance wave approach to calculate the transmission between two plates and Hambric et al. [24] compared various models of a T-joint and showed that at higher frequencies the finite dimensions of the joint become important and should be included in the model.

The waveguides in the reviewed literature were mostly isotropic and the joints were often idealised (e.g. point connections). For such cases, analytical models can be developed both for wave propagation and the continuity and equilibrium conditions at the joint. However, for complicated waveguides and/or complicated joints, finding the reflection and transmission matrices of the joints using a purely analytical formulation can be extremely difficult at best, so that a numerical approach might be required. In this paper, a hybrid approach for calculating the reflection and transmission matrices of joints is presented. Joints are modelled using standard FE methods whereas the waveguides are modelled using the wave and finite element (WFE) method. This relies on post-processing a standard FE model of a small segment of each waveguide using periodic structure theory. The degrees of freedom at the interfaces of the WFE and FE models of the waveguides and the joint are compatible. The models are coupled to yield the reflection and transmission matrices of the joint. These can then be used to obtain the response of the whole structure. The approach is similar to that described by Doyle [10] in which a joint is modelled in FE and coupled with waveguides which are modelled by spectral elements formed from analytical models. However, this suffers from two possible problems: first if the waveguide has a complicated construction (e.g. a laminate or a solid or thin-walled section for which higher order wavemodes are important), then finding the spectral element analytically can be very difficult. Secondly, in such an

approach, the models of the waveguides and the joint are not compatible at their interface. While the models can be coupled (e.g. [10] or by the use of Lagrange multipliers), there is a discontinuity in the models, which results in spurious, although often small, additional scattering effects at the interface [25, 26]. In the approach described here, these issues are not relevant: the WFE method is used to determine the wavemodes so that the complexity of the construction is arbitrary, and the interfaces between the joint and the waveguides have compatible meshes.

The WFE method has been used to study the free [27] and forced [28] vibration of waveguides. The WFE method for waveguides has also been used to study thin-walled structures [29], laminated plates [27] and fluid filled pipes [30, 31]. It has also been extended to two dimensional plane [32] and cylindrical structures [33].

The remainder of this paper is organised as follows. In Section 2, the WFE method will be reviewed briefly. The scattering properties of joints will be obtained in Section 3. The response of finite waveguide structure is discussed in Section 4 and numerical examples are presented in Section 5. Finally, conclusions for this work are presented in Section 6.

2 REVIEW OF THE WFE METHOD

The WFE method is a method for modelling one and two-dimensional piecewise homogenous/periodic structures. Waki et al. [34] provides a comprehensive review of the method.

2.1 Free wave propagation and the wave basis

The WFE method starts with obtaining the FE model of a small segment of the waveguide (shown in Figure 1) using any FE package with the only constraint being that the nodes and their degrees of freedom (DOF)s are ordered identically on the left and right sides of the segment.

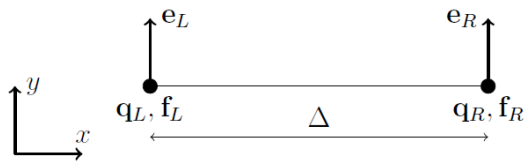

Figure 1 FE model of a segment of a waveguide with the waveguide's local coordinate system.

Internal nodes can be eliminated via dynamic condensation [34]. Time harmonic dependence of the form $e^{i\omega t}$ is assumed throughout this work and is suppressed for brevity. The governing equation of the segment of Figure 1 is

$$\mathbf{D}\,\mathbf{q} = \mathbf{f} + \mathbf{e} \quad \text{with} \quad \mathbf{D}=\mathbf{K} + i\omega\mathbf{C} - \omega^2\mathbf{M}\,, \tag{1}$$

where $\mathbf{q}$, $\mathbf{f}$ and $\mathbf{e}$ are $(2n\times 1)$ vectors of nodal DOFs, internal and external nodal forces respectively, $\mathbf{D}$, $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{K}$ are the dynamic stiffness, mass, viscous damping and stiffness matrices of the segment and n is the number of DOFs at each side of the segment. The dynamic stiffness matrix can be partitioned to reflect the influence of the left and right nodes of the segment, and thus Equation (1) can be expressed as

$$\begin{bmatrix} \mathbf{D}_{LL} & \mathbf{D}_{LR} \\ \mathbf{D}_{RL} & \mathbf{D}_{RR} \end{bmatrix} \begin{Bmatrix} \mathbf{q}_L \\ \mathbf{q}_R \end{Bmatrix} = \begin{Bmatrix} \mathbf{f}_L \\ \mathbf{f}_R \end{Bmatrix} + \begin{Bmatrix} \mathbf{e}_L \\ \mathbf{e}_R \end{Bmatrix}, \tag{2}$$

where the subscripts L and R denote the left and right sides of the segment, respectively. When a wave propagates freely through the waveguide, the propagation constant $\lambda = e^{-ik\Delta}$ relates the right and left nodal DOFs and forces by

$$\mathbf{q}_R = \lambda\,\mathbf{q}_L \quad , \quad \mathbf{f}_R = -\lambda\,\mathbf{f}_L\,. \tag{3}$$

Substituting Equation (3) into Equation (2) yields an eigenvalue problem for λ which upon solving yields the wave properties of the waveguide [34]. The eigenvalues occur in reciprocal pairs as λ_j^+ and $\lambda_j^- = 1/\lambda_j^+$ [35, 36], with wavenumbers k_j^+ and $k_j^- = -k_j^+$, corresponding to positive- and negative-going waves respectively. Associated with these eigenvalues are the positive- and negative-going eigenvectors $\boldsymbol{\varphi}_j^+$ and $\boldsymbol{\varphi}_j^-$ respectively, which are also called the wavemodes. Every wavemode can be partitioned as

$$\boldsymbol{\varphi}_j = \begin{Bmatrix} \boldsymbol{\varphi}_q \\ \boldsymbol{\varphi}_f \end{Bmatrix}_j\,. \tag{4}$$

The wave modes can be grouped into

$$\boldsymbol{\Phi}^{\pm} = \begin{bmatrix} \boldsymbol{\varphi}_1^{\pm} & \cdots & \boldsymbol{\varphi}_n^{\pm} \end{bmatrix}, \boldsymbol{\Phi} = \begin{bmatrix} \boldsymbol{\Phi}^{+} & \boldsymbol{\Phi}^{-} \end{bmatrix}. \tag{5}$$

The left wavemodes can also be obtained. These are $(1\times 2n)$ vectors which can be partitioned as

$$\boldsymbol{\psi}_j = \begin{bmatrix} \boldsymbol{\psi}_f & \boldsymbol{\psi}_q \end{bmatrix}_j, \tag{6}$$

and further grouped into

$$\boldsymbol{\Psi} = \begin{bmatrix} \boldsymbol{\Psi}^{+} \\ \boldsymbol{\Psi}^{-} \end{bmatrix}, \text{where } \boldsymbol{\Psi}^{\pm} = \begin{bmatrix} \boldsymbol{\psi}_1^{\pm} \\ \vdots \\ \boldsymbol{\psi}_n^{\pm} \end{bmatrix}, \quad j = 1,\ldots,n\,. \tag{7}$$

The left and right wavemodes can be normalised [34] such that

$$\boldsymbol{\Psi}^{\pm}\boldsymbol{\Phi}^{\pm} = \mathbf{I} \quad \text{yielding} \quad \boldsymbol{\Psi}\,\mathbf{T}\boldsymbol{\Phi} = \text{diag}\left(\lambda_j\right), \tag{8}$$

where diag(.) represents a diagonal matrix. The partitions of the left and right eigenvectors can be used to form the matrices

$$\boldsymbol{\Phi}_q^{\pm} = \begin{bmatrix} \boldsymbol{\varphi}_{q,1}^{\pm} & \cdots & \boldsymbol{\varphi}_{q,n}^{\pm} \end{bmatrix}, \boldsymbol{\Psi}_q^{\pm} = \begin{bmatrix} \boldsymbol{\psi}_{q,1}^{\pm} \\ \vdots \\ \boldsymbol{\psi}_{q,n}^{\pm} \end{bmatrix}, \quad j = 1,\ldots,n\,. \tag{9}$$

Similar expressions hold for $\boldsymbol{\Phi}_f^{\pm}$ and $\boldsymbol{\Psi}_f^{\pm}$. These matrices, together with the orthogonality relations of Equation (8), define transformations between the physical domain, where the motion is described in terms of **q** and **f**, and the wave domain, where the motion is described in terms of waves of

amplitudes $\mathbf{a}^+$ and $\mathbf{a}^-$ travelling in the positive and negative directions respectively. Specifically,

$$\begin{Bmatrix} \mathbf{q}_L \\ \mathbf{f}_L \end{Bmatrix} = \mathbf{\Phi}\,\mathbf{a} \quad , \quad \mathbf{a} = \begin{Bmatrix} \mathbf{a}^+ \\ \mathbf{a}^- \end{Bmatrix}. \tag{10}$$

In practice, as in modal analysis, only m pairs of (positive- and negative-going) waves might be retained, so that $\mathbf{\Phi}^{\pm}_{q,f}$ and $\mathbf{\Psi}^{\pm}_{q,f}$ are $n \times m$ and $m \times n$ matrices respectively. The number retained can be different at different frequencies [37].

2.2 Forced response

A point excitation $\mathbf{f}_e$ will generate positive- and negative-going waves of amplitudes $\mathbf{a}^+$ and $\mathbf{a}^-$ which will propagate away from the excitation point in the right and left directions respectively. The amplitudes of these waves can be obtained as [34]

$$\mathbf{a}^+ = \mathbf{\Psi}^+_q \mathbf{f}_e \quad , \quad \mathbf{a}^- = -\mathbf{\Psi}^-_q \mathbf{f}_e \,. \tag{11}$$

In the case of spatially distributed loads, the amplitudes of the excited waves can be obtained analytically by using contour integration techniques along with the WFE model [38].

3 SCATTERING PROPERTIES OF JOINTS

The reflection and transmission coefficients of joints in waveguide structures will now be evaluated using a hybrid FE/WFE approach. First, point joints will be treated, followed by the case of joints of finite dimensions.

3.1 Point joints

Figure 2 shows p waveguides with zero-dimensional cross-sections connected at a point joint[1]. For the *j*-th waveguide, the local coordinate system is defined such that the waveguide's axis is directed towards the joint. The rotation matrix $\mathbf{R}_j$ transforms the DOFs from the local (x_j, y_j) to the global (X, Y) coordinate system.

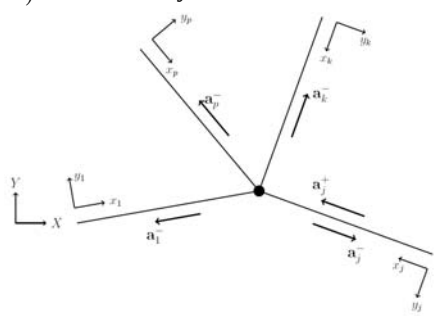

Figure 2 Waveguides connected at a point joint with incident positive-going waves.

The WFE model of each waveguide is obtained in its local coordinate system (x_j, y_j). The incident, reflected and transmitted waves are related through the scattering matrix as

$$\mathbf{a}^-_k = \mathbf{s}_{kj}\mathbf{a}^+_j \quad j,k = 1,\ldots,p \,, \tag{12}$$

where $\mathbf{s}_{jj} = \mathbf{r}_{jj}$ is the reflection matrix of the joint at the *j*-th waveguide, and $\mathbf{s}_{kj} = \mathbf{t}_{kj}$ (for $j \neq k$) is the transmission matrix of the joint from the *j*-th to the *k*-th waveguides.

It is assumed that the DOFs of the waveguides are compatible and all the waveguides are modelled using the same number of DOFs. If the DOFs of the waveguides are incompatible, the interface can then be modelled using full FE and the results of the next section can be used.

Using the continuity and equilibrium conditions at the joint, the scattering matrix can be obtained as [37]

$$\mathbf{s} = -\left[\mathbf{C}^-\right]^{-1}\mathbf{C}^+ \,, \tag{13}$$

where

$$\mathbf{C}^{\pm} = \begin{bmatrix} \mathbf{R}_1\mathbf{\Phi}^{\pm}_{f,1} - \mathbf{D}_J\mathbf{R}_1\mathbf{\Phi}^{\pm}_{q,1} & \mathbf{R}_2\mathbf{\Phi}^{\pm}_{f,2} & \cdots & \mathbf{R}_p\mathbf{\Phi}^{\pm}_{f,p} \\ \mathbf{R}_1\mathbf{\Phi}^{\pm}_{q,1} & -\mathbf{R}_2\mathbf{\Phi}^{\pm}_{q,2} & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{R}_1\mathbf{\Phi}^{\pm}_{q,1} & \mathbf{0} & \cdots & -\mathbf{R}_p\mathbf{\Phi}^{\pm}_{q,p} \end{bmatrix}. \tag{14}$$

Some waveguides might be connected such that the outgoing waves from one joint are incident on a second join, Figure 3.

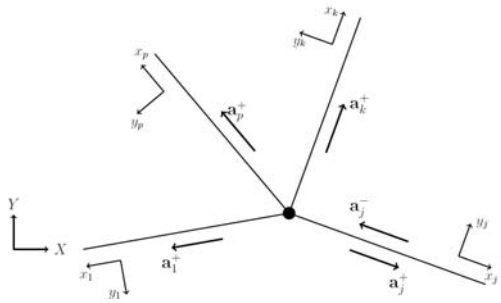

Figure 3 Waveguides connected at a point joint with incident negative-going waves.

In this case, the scattering matrix becomes

$$\tilde{\mathbf{s}} = -\left[\mathbf{C}^+\right]^{-1}\mathbf{C}^- \,. \tag{15}$$

If only $m_j < n$ pairs of (positive- and negative-going) waves are retained for the WFE model of the *j*-th waveguide, then $\mathbf{C}^{\pm}$ is not invertible. In this case, pseudo-inversion should be used, and the scattering matrix becomes

$$\mathbf{s} = -\left[\mathbf{G}^-\mathbf{C}^-\right]^{-1}\mathbf{G}^-\mathbf{C}^+ \quad \text{or} \quad \tilde{\mathbf{s}} = -\left[\mathbf{G}^+\mathbf{C}^+\right]^{-1}\mathbf{G}^+\mathbf{C}^- \,, \tag{16}$$

where

$$\mathbf{G}^{\pm} = \begin{bmatrix} \mathbf{\Psi}^{\pm}_{f,1}\mathbf{R}^T_1 & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & \mathbf{\Psi}^{\pm}_{f,p}\mathbf{R}^T_p \end{bmatrix}. \tag{17}$$

[1] For simplicity the waveguides are assumed to lie in a plane: if this is not the case, the only difference is related to the transformation matrices $\mathbf{R}_j$ of the waveguides.

3.2 *Finite joints*

Next, the reflection and transmission properties of joints of finite size will be considered. Suppose p waveguides are attached at a joint, Figure 4. The waveguides are modelled using the WFE method of Section 2, and the joint is modelled using standard FE. The WFE model of each waveguide is obtained in its local coordinate system $\left(x_j, y_j\right)$ using n_j DOFs (on each side of the waveguide segment) and m_j pairs of positive- and negative-going waves in the WFE model. It is assumed that the interface DOFs of the joint are compatible with those of the WFE models.

Time harmonic behaviour of the joint is described through

$$\begin{bmatrix} \tilde{\mathbf{D}}_{ii} & \tilde{\mathbf{D}}_{in} \\ \tilde{\mathbf{D}}_{ni} & \tilde{\mathbf{D}}_{nn} \end{bmatrix} \begin{Bmatrix} \mathbf{Q}_i \\ \mathbf{Q}_n \end{Bmatrix} = \begin{Bmatrix} \mathbf{F}_i \\ \mathbf{F}_n \end{Bmatrix}, \tag{18}$$

where $\mathbf{Q}$ and $\mathbf{F}$ are vectors of DOFs and internal nodal forces represented in the global coordinate system $\left(X, Y\right)$ and the subscripts i and n represent interface and non-interface nodes, respectively. Since it is assumed that no external forces are applied at the non-interface nodes ($\mathbf{F}_n = \mathbf{0}$), the FE model of the joint can be condensed as

$$\begin{aligned} &\mathbf{D}_{ii}\mathbf{Q}_i = \mathbf{F}_i, \\ &\mathbf{D}_{ii} = \tilde{\mathbf{D}}_{ii} - \tilde{\mathbf{D}}_{in}\tilde{\mathbf{D}}_{nn}^{-1}\tilde{\mathbf{D}}_{ni} \text{ and } \mathbf{Q}_n = -\tilde{\mathbf{D}}_{nn}^{-1}\tilde{\mathbf{D}}_{ni}\mathbf{Q}_i \end{aligned} \tag{19}$$

The above condensation is the simplest; other approaches can be used for enhanced computational speed and better accuracy [10, 25]. Moreover, the interface nodes can be separated into those corresponding to each waveguide, Figure 4.

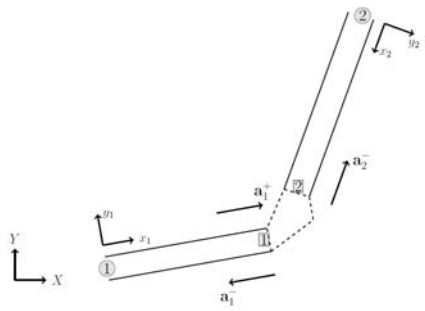

Figure 4 Schematic of two waveguides attached at a joint with incident, reflected and transmitted waves (the number of waveguides conned to the joint could be arbitrary but here two are shown for simplicity).

Vectors $\mathbf{q}$ and $\mathbf{f}$ and matrices $\mathbf{\Phi}_{q,f}^{\pm}$, $\mathbf{\Psi}_{q,f}^{\pm}$ and $\mathbf{R}$ are now defined by concatenating the relevant vectors and matrices for the individual waveguides (for $\mathbf{\Phi}_{q,f}^{\pm}$, $\mathbf{\Psi}_{q,f}^{\pm}$ and $\mathbf{R}$, the matrices are block-diagonal). The continuity and equilibrium conditions can be used to formulate the scattering matrix as [37]

$$\mathbf{s} = -\left[\mathbf{R}\mathbf{\Phi}_f^- - \mathbf{D}_{ii}\mathbf{R}\mathbf{\Phi}_q^-\right]^{-1}\left[\mathbf{R}\mathbf{\Phi}_f^+ - \mathbf{D}_{ii}\mathbf{R}\mathbf{\Phi}_q^+\right]. \tag{20}$$

If the outgoing waves are incident upon the joint, then the scattering matrix is

$$\tilde{\mathbf{s}} = -\left[\mathbf{R}\mathbf{\Phi}_f^+ - \mathbf{D}_{ii}\mathbf{R}\mathbf{\Phi}_q^+\right]^{-1}\left[\mathbf{R}\mathbf{\Phi}_f^- - \mathbf{D}_{ii}\mathbf{R}\mathbf{\Phi}_q^-\right]. \tag{21}$$

Once again, if a reduced number of waves is retained, then pseudo-inversion is required, and the scattering matrix becomes

$$\mathbf{s} = -\left[\mathbf{\Psi}_f^-\left(\mathbf{R}\mathbf{\Phi}_f^- - \mathbf{D}_{ii}\mathbf{R}\mathbf{\Phi}_q^-\right)\right]^{-1}\mathbf{\Psi}_f^-\left[\mathbf{R}\mathbf{\Phi}_f^+ - \mathbf{D}_{ii}\mathbf{R}\mathbf{\Phi}_q^+\right], \tag{22}$$

or

$$\tilde{\mathbf{s}} = -\left[\mathbf{\Psi}_f^+\left(\mathbf{R}\mathbf{\Phi}_f^+ - \mathbf{D}_{ii}\mathbf{R}\mathbf{\Phi}_q^+\right)\right]^{-1}\mathbf{\Psi}_f^+\left[\mathbf{R}\mathbf{\Phi}_f^- - \mathbf{D}_{ii}\mathbf{R}\mathbf{\Phi}_q^-\right], \tag{23}$$

4 FINITE WAVEGUIDE STRUCTURES

The WFE models (developed in Section 2) and the scattering properties of the joints (obtained in Section 3) can be used to obtain the response of waveguide structures. Each waveguide is modelled using the WFE approach using its local coordinate system. The calculation of the scattering matrix, $\mathbf{s}^{jk}$ of the joint connecting the *j*-th and *k*-th waveguides, will depend on the orientation of the axes of these waveguide.

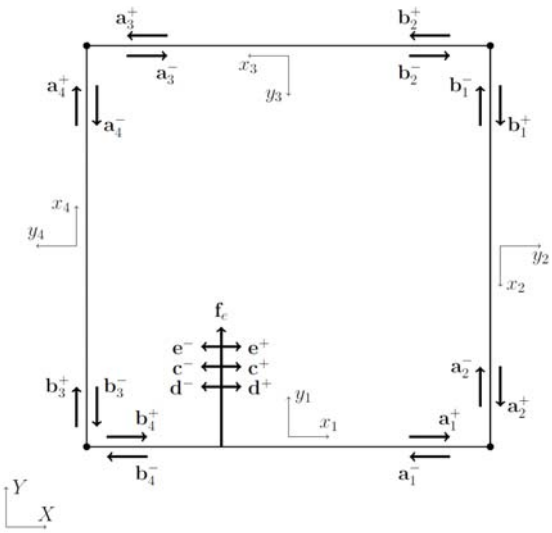

Figure 5 A closed waveguide structure of four waveguides. The orientation of the waveguides can be arbitrary, and any waveguide can be excited at any point.

Figure 5 shows an example of a closed waveguide structure comprising four waveguides, each of length $L_j\left(j = 1, \cdots, 4\right)$. A point excitation is applied on waveguide 1, which generates waves of amplitudes $\mathbf{e}^{\pm}$. The amplitudes of these waves can be obtained directly from Equation (11). As for the amplitudes of the travelling waves, their amplitudes can be related to the excited waves as

$$\mathbf{c}^+ = \mathbf{e}^+ + \mathbf{d}^+ \quad , \quad \mathbf{d}^- = \mathbf{e}^- + \mathbf{c}^-. \tag{24}$$

The amplitudes of the incident and scattered waves are related through the scattering matrices as

$$\mathbf{a}^- = \begin{bmatrix} \mathbf{s}^{12} & \\ & \mathbf{s}^{34} \end{bmatrix}\mathbf{a}^+ \quad , \quad \mathbf{b}^+ = \begin{bmatrix} \tilde{\mathbf{s}}^{23} & \\ & \tilde{\mathbf{s}}^{41} \end{bmatrix}\mathbf{b}^-, \tag{25}$$

where

$$\mathbf{a}^{\pm} = \begin{Bmatrix} \mathbf{a}_1^{\pm} \\ \vdots \\ \mathbf{a}_4^{\pm} \end{Bmatrix} \quad , \quad \mathbf{b}^{\pm} = \begin{Bmatrix} \mathbf{b}_1^{\pm} \\ \vdots \\ \mathbf{b}_4^{\pm} \end{Bmatrix}. \tag{26}$$

Finally, in the unexcited waveguides (2, 3 and 4), the wave amplitudes are related through

$$\begin{Bmatrix} \mathbf{a}_{j+1}^{+} \\ \mathbf{b}_{j}^{-} \end{Bmatrix} = \begin{bmatrix} \boldsymbol{\tau}_j\left(L_j\right) & \\ & \boldsymbol{\tau}_j\left(L_j\right) \end{bmatrix} \begin{Bmatrix} \mathbf{b}_{j}^{+} \\ \mathbf{a}_{j+1}^{-} \end{Bmatrix} \quad , \quad j = 2,3 . \tag{27}$$

where

$$\boldsymbol{\tau}_j\left(x\right) = \mathrm{diag}\left(e^{-ik_{1,j}^{+}x}, \cdots, e^{-ik_{m_j,j}^{+}x}\right) \tag{28}$$

is the wave propagation matrix that gives the amplitudes of the waves after travelling a distance x in the j-th waveguide. For the excited waveguide (1 in Figure 5), these relations are

$$\begin{aligned} \mathbf{b}_4^{-} &= \boldsymbol{\tau}_1\left(x_e\right)\mathbf{d}^{-}, \mathbf{d}^{+} = \boldsymbol{\tau}_1\left(x_e\right)\mathbf{b}_4^{+}, \\ \mathbf{c}^{-} &= \boldsymbol{\tau}_1\left(L - x_e\right)\mathbf{a}_1^{-}, \mathbf{a}_1^{+} = \boldsymbol{\tau}_1\left(L - x_e\right)\mathbf{c}^{+}. \end{aligned} \tag{29}$$

Equations (24) through (29) can be solved to yield the amplitudes of the waves travelling through the whole structure. Then, the response at any point of the waveguide can be obtained straightforwardly [34].

5 NUMERICAL EXAMPLES

In this section, three numerical examples are presented. Analytical solutions for the first example, four Euler-Bernoulli beams forming a rectangle, can be obtained. The second example is of four simply supported plate strips also forming a rectangle and connected at line joints. The last example is of a cross-section of a train cabin where the hybrid approach of Section 3.2 proves to be useful. In the following, all properties and dimensions are in SI units.

5.1 Beam structure

In this example, a rectangular beam structure is considered, Figure 5. The point joining every two beams has a mass of 0.25 kg. The beams are made of mild steel with a density $\rho = 7800$, modulus of elasticity $E = 200 \times 10^9$, Poisson ratio $\nu = 0.3$, and structural damping $\eta = 0.03$. The four beams are identical and the width, thickness and length are $b = 0.03$, $h = 0.01$ and $L = 0.6$ respectively. The beams are slender, so Euler-Bernoulli beam theory is used (but other theories can be equally used).

Since the beams are identical, then it is sufficient to obtain a single WFE model. This is achieved via the FE model of a single BEAM3 element of ANSYS® ($\Delta = 0.01$).BEAM3 is a two-noded element with 3 DOFs per node. The approach presented in Section 3 and 4 is used to calculate the scattering properties of the joints and subsequently find the response of the structure. For comparison, the response can be obtained analytically using a dynamic stiffness matrix approach, e.g. [39]. A unit amplitude force in the $y-$direction is applied at the middle of the lower beam. Figure 6 shows that the WFE/FE results are in good agreement with the analytical results.

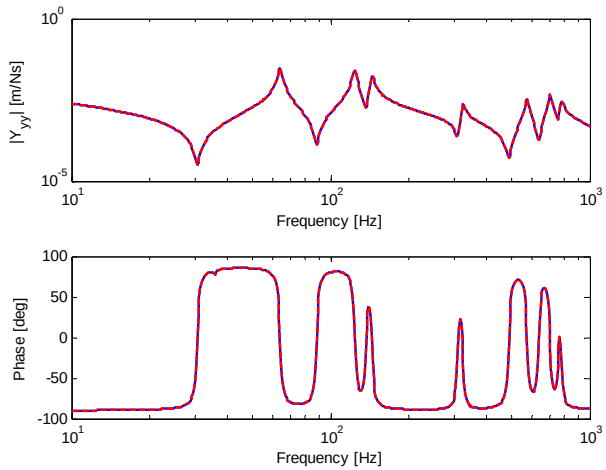

Figure 6 Input mobility in the $y-$direction at $x = 0.3$ m on the lower beam: — analytical, -- WFE/FE.

5.2 Plate structure

Consider four plates that are connected at their edges to form a rectangular frame as in Figure 5. The frame is simply supported at all the other edges. The plates are made of mild steel (material properties are given in the first example). The plates have the same length $L = 0.6$, are 18 cm wide and 1.8 mm thick.

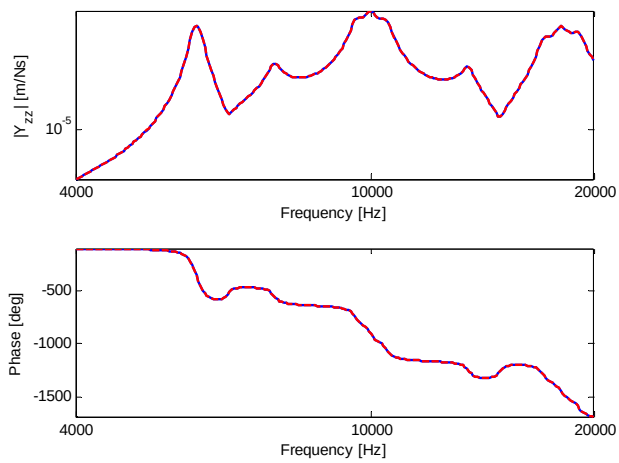

Figure 7 Transfer mobility in the $z-$direction of the upper plate: — full FE, -- WFE/FE.

Since the four plates are identical, then one WFE model is sufficient. This was obtained using a segment (with $\Delta = 0.02$) of nine SHELL63 elements of ANSYS®. SHELL63 is a four-noded element with six DOFs per node. The segment has 20 nodes with a total of 120 DOFs. However, five DOFs are removed from each of the four boundary nodes. Thus, the system to be processed as shown in Section 2 has 100 DOFs.

For comparison, the whole structure is modelled with FE using elements of the same size. The full FE model has 5992 DOFs. A unit amplitude force is applied normal to the lower plate at $(0.3, 0.08)$, and the response of the upper plate is observed at $(0.5, 0.08)$. The transfer mobility is shown in

Figure 7. The WFE/FE results are in good agreement with the full FE results.

5.3 *Two-dimensional cross-section of train cabin*

Train floor panels comprise panels with a truss-like core. These panels are usually made of aluminium with density $\rho = 2700$, modulus of elasticity $E = 72 \times 10^9$, Poisson ratio $\nu = 0.34$, and structural damping $\eta = 0.03$. Such a structure is periodic in the $x-$direction and homogeneous in the $y-$direction, Figure 8. The walls and ceiling of the train cabin are often of a similar construction.

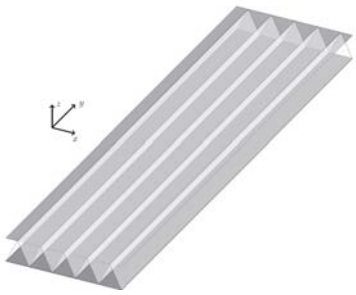

Figure 8 Schematic of a train floor.

To illustrate the applicability of the developed approach for studying the behaviour of such structures, the model of a section in the $y-$direction will be developed, based on a plain strain assumption. Thus, the section can be modelled using beam elements of an equivalent modulus of elasticity given as

$$E_{Equivelant} = \frac{E}{\left(1-\nu^2\right)}. \tag{30}$$

The results obtained will be valid per unit width. For simplicity (and without any loss of generality), the walls and ceiling are assumed to be identical to the floor with the joints (also being identical) displayed in Figure 9. Each of the floor and ceiling comprises 25 cells, whereas each of the two walls comprises 20 cells.

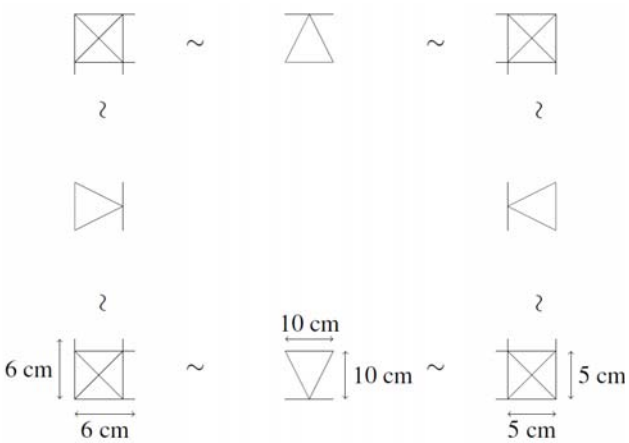

Figure 9 Simplified cross-section of a train cabin.

The WFE model of the four waveguides (ceiling, floor, and two walls) can be realised by modelling a single cell using BEAM3 elements of ANSYS® (with an element length of 1cm where possible). The FE model of the cell has 102 DOFs, but the internal DOFs can be condensed bringing down the size of the FE model of the cell to 12 DOFs. The joints (also made of aluminium) are modelled using FE (with the same element length), and this results in an FE model of 111 DOFs. At low frequencies, the truss-cored panel behaves like an orthotropic plate with an axial, shear and bending wave. At higher frequencies, above 1100 Hz, the periodicity of the truss-core in the x-direction becomes apparent and higher-order waves cut-on. These comprise motion in the face-plates and core webs which is difficult to divide into simple categories.

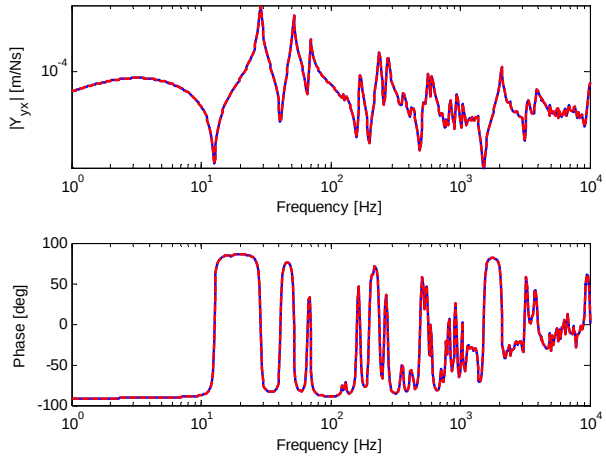

Figure 10 Transfer mobility of the simplified train cabin: — full FE, -- WFE/FE.

The approach presented in Section 4 is now used to compute the scattering properties of the joints. Then, a point force is applied to the lower waveguide at 1.3 m from its left end. The response is observed at 1.3 m from the bottom end of the right waveguide. For comparison, the whole cross-section is modelled using FE with elements of the same size. This yields a model of 4560 DOFs. The results of the WFE/FE results, Figure 10, are in very good agreement with the full FE results.

6 CONCLUSIONS

In this paper a hybrid wave finite element/finite element (WFE/FE) approach for calculating the scattering properties of joints is used to obtain the vibrational behaviour of a waveguide structure. The waveguides are modelled using the WFE method where the FE model of a small segment of the waveguide is post-processed using periodic structure theory to obtain the wave characteristics of the waveguide. The joints are modelled using FE, and the WFE and FE models are coupled to calculate the scattering properties of the joints. Thus, the vibrational behaviour of the structure is described in the wave domain which has many advantages. First, the resulting models are smaller in size which becomes useful at higher frequencies where FE models become impractical. On the other hand, the knowledge of wave properties is useful for other applications (statistical energy analysis, structure-borne sound, disturbance propagation, etc). Numerical examples were presented to illustrate the approach.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the financial support provided by the Engineering and Physical Sciences Research Council under grant number EP/F069391/1 and the European Commission in the context of the collaborative project "Mid-Mod: Mid-frequency vibro-acoustic modelling tools-innovative CAE methodologies to strengthen European competitiveness" (Grant agreement number: 218508).

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